

TEXT ANONYMIZATION · PRIVACY ENGINEERING

Making Your Data Shareable Without Making It Useless

DP-MLM-X: Risk-aware differential privacy for text records

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BEFORE WE START

Six questions.

Same person.

Each question hides a little more.
How far can you follow?

Q1 · Trivial

Q2 · Context

Q3 · Guess

Q4 · Background

Q5 · Harder

Q6 · Impossible

| Who is this?

Q1 **Trivial — name visible**

Jane Doe was born on **4 February 1990** in **Stuttgart**. She completed her studies in **computer science** at **TU Berlin** and currently works as a **senior data engineer** at **Allianz SE**. She has been based in Munich since 2019.

Name, dates, employer all present. Takes 2 seconds.

Q2 **Context clues remain**

J.D. was born on **[DATE]** in **Stuttgart**. She studied **computer science** at **TU Berlin** and works as a **senior data engineer** at **Allianz SE**. Based in Munich since 2019.

Initials + employer make this solvable for most people.

| Getting harder.

Q3 **No direct clues**

[PERSON] was born in [YEAR] in [CITY]. She studied computer engineering and currently works as a data specialist in southern Germany. She has worked in insurance technology for several years.

No name. You'd need domain knowledge to narrow this down.

Q4 **Background knowledge helps**

[PERSON] was born in [LOCATION]. They completed graduate studies at a technical university and joined a financial services firm as an engineering lead. Currently based in a major German city.

Hint: this record is from a dataset of German engineering graduates.

| Final two.

Q5

Obscure background knowledge

[PERSON] was born in [YEAR] in [COUNTRY]. They completed a degree in [FIELD] and transitioned into [INDUSTRY] around [YEAR]. Currently employed at a mid-size firm.

Hint: record from a European professional dataset, graduation 1990–2000.

Q6

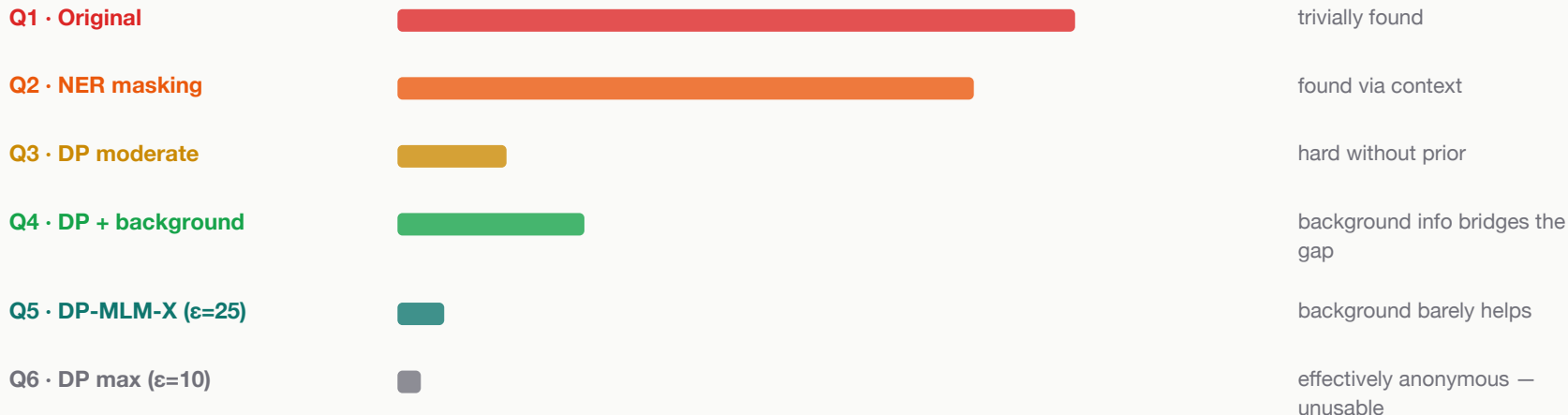
Random guess only

[PERSON] was embargoed on [DATE] in [CITY]. She performed numerical calibration at [INSTITUTION] and currently monitors regulatory compliance at a decentralized infrastructure unit. Relocated to an undisclosed region in 2021.

Mathematically private. Essentially useless.

| Here's what a re-ID model scored.

MRR = Mean Reciprocal Rank. A score of 1.0 = always ranked 1 out of 2,419 candidates. A score near 0 = effectively hidden in the crowd.



Data: TAB dataset · 2,419 candidate pool · white-box re-ID model. Q4 uses black-box (background knowledge) evaluation.

| Why does hiding the name not work?

Compliance teams mask **names, dates, locations** — the easy targets.

But a re-ID model doesn't read like a human. It looks for the **combination** of signals that together uniquely identify someone:

- ◆ occupation + seniority level
- ◆ employer / industry / company size
- ◆ city + timeline of moves
- ◆ language patterns and writing style

Remove the name and these still form a **semantic fingerprint**. In our experiments, Presidio (NER masking) reduces MRR from 0.87 → 0.74. The person is still ranked first 74% of the time.

EXAMPLE: Q1 → Q2 comparison

Q1 (MRR = 0.87) Jane Doe, Stuttgart, born 1990, senior data engineer, Allianz SE

Q2 (MRR = 0.74) [PERSON], [CITY], born [YEAR], senior data engineer, Allianz SE

← Employer + seniority alone narrow it to 5 candidates in any realistic corporate dataset.

I Differential privacy: the idea in one sentence.

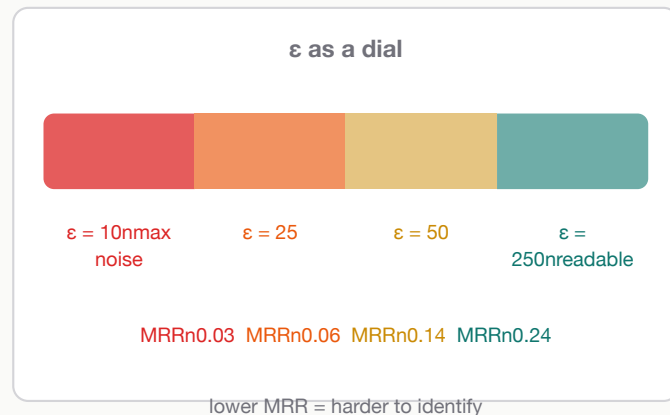
The guarantee: if you change any single token in the input, the output distribution barely changes — by at most a factor of e^ϵ in probability.

The knob: ϵ (epsilon).

-  small $\epsilon \rightarrow$ strong privacy, more noise
-  large $\epsilon \rightarrow$ weaker privacy, readable text

The mechanism: each token gets replaced by a randomly sampled alternative, where the sampling distribution is calibrated so that no output reveals which word was really there.

This is the gap between Q2 and Q3. Q3 uses $\epsilon = 50$. Q6 uses $\epsilon = 10$ — so noisy the text falls apart.



| Not all tokens need the same protection.

Uniform DP-MLM treats every word identically. “the” gets the same noise budget as “Deutsche Bahn.”

DP-MLM-X uses SHAP (a standard feature-attribution method) to score each token’s marginal contribution to re-identification risk. Then it **concentrates the privacy budget on the dangerous tokens** and relaxes it for the harmless ones.

Same total budget. Smarter allocation.

At $\epsilon = 50$:

Uniform DP-MLM

MRR = 0.449 (found 45% of the time)

DP-MLM-X

MRR = 0.138 (found 14% of the time)

← 3× stronger privacy. Same ϵ . Better readability too.

SHAP risk scores per token

John Smith was born on **12 March 1987** in **Munich**. He worked as a **senior engineer** at **Deutsche Bahn** from 2014 to 2019.

-  High risk — gets maximum noise
-  Medium risk — moderate noise
-  Low risk — barely touched

Budget concentrates on **names, dates, employers**. “was”, “in”, “from” consume almost nothing.

| Where each method lives.

Privacy (P) = MRR, lower is better. Utility (U) = text quality score, higher is better. Data: TAB dataset.

Privacy (MRR, lower = better) × Utility (higher = better) — all methods

DP-MLM-X consistently sits in the top-right quadrant: lower MRR (stronger privacy) at the same or higher utility than uniform DP-MLM. Relative gain (RG = combined score) is highest for DP-MLM-X at $\epsilon \geq 100$ on both datasets.

| The numbers at $\epsilon = 50$.

TAB dataset · $\epsilon = 50$ · all DP methods share the same privacy budget

Method	Privacy (MRR↓)	Utility (U↑)	Relative Gain↑	Time (s)
Original (no anon.)	0.869	0.988	—	—
Presidio (NER)	0.740	0.951	0.110	0.1
SpaCy (NER)	0.396	0.562	0.113	0.0
DP-Paraphrase	0.041	0.474	0.433	14
DP-MLM (uniform)	0.449	0.897	0.391	80
DP-MLM-X (ours)	0.138	0.616	0.465	42

Key At $\epsilon=50$, DP-MLM-X achieves **3× lower MRR** than uniform DP-MLM while running **2× faster**. Utility drops (0.897→0.616) because risky tokens are more aggressively replaced — but the text remains coherent and downstream tasks still work.

| Two knobs. Calibrate once. Deploy.

ϵ — Privacy budget

Set this to match your compliance target.

- Strict (GDPR Art. 89 spirit): $\epsilon = 10\text{--}25$
- Balanced: $\epsilon = 50\text{--}100$
- Light touch (internal data): $\epsilon = 250$

Calibrate against a **held-out validation set**: measure MRR vs. utility at each ϵ , pick the operating point that satisfies your SLA.

τ — Risk temperature

Controls how aggressively budget concentrates on risky tokens.

- $\tau = 0.1$: maximum concentration, best privacy efficiency, text can be choppy
- $\tau = 1.0$: moderate, good balance of privacy and readability
- $\tau = 2.0+$: near-uniform (behaves like standard DP-MLM)

In experiments, $\tau = 0.1$ gives **21× budget savings** at the same 90% anonymity coverage vs. uniform DP-MLM.

| What it takes to run this.

Dependencies

```
pip install transformers datasets shap torch
```

Core API

```
from dp.loaders import get_adapter

adapter = get_adapter(
    "tab",
    data="tab",
    data_in="path/to/records",
)

# anonymize one record
record = next(adapter.iter_records())
output = anonymize(record, epsilon=50, tau=0.1)
```

Typical latency (TAB dataset, GPU)

Method	Time / record	ϵ range
DP-MLM (uniform)	80 s	10–250
DP-MLM-X (full)	42 s	50–250
ρ -DP-MLM-X ($\rho=0.5$)	27 s	10
k-DP-MLM-X ($k=10$)	47 s	10

Notes

- SHAP pass is computed **once** per record and cached
- GPU required for practical throughput; CPU works for demos
- Batch processing supported via `--data / --data_in` CLI flags

| Decision guide.

Situation	Recommended approach	Privacy level	Utility trade-off
Internal analytics, no PII regulation	Skip — use the raw data	None needed	Full utility
Compliance audit, names must go	NER masking (Presidio)	Weak (MRR ≈ 0.74)	Minimal loss
GDPR-adjacent sharing, text quality matters	DP-MLM-X, $\epsilon = 100\text{--}250$	Moderate (MRR $0.20\text{--}0.24$)	Good (U $\approx 0.78\text{--}0.83$)
Research dataset release, strong guarantee needed	DP-MLM-X, $\epsilon = 25\text{--}50$	Strong (MRR $0.06\text{--}0.14$)	Moderate (U $\approx 0.42\text{--}0.62$)
Maximum privacy, readability secondary	ρ -DP-MLM-X, $\rho = 0.1$, $\epsilon = 10$	Very strong (MRR ≈ 0.04)	Lower (U ≈ 0.38)

TAKEAWAYS

01

Name masking isn't enough

Semantic fingerprints survive NER.
You need to randomize the content,
not just the labels.

02

Spend the budget where it counts

Risk-weighted DP gives 3× stronger
privacy than uniform DP-MLM at the
same ϵ — without spending more.

03

Two knobs, one decision

Set ϵ to your compliance target, $\tau = 0.1$ for max savings. Calibrate once
on held-out data. Deploy.

Questions? The code and paper are both open.